

1. **What are the main characteristics of hollow concrete blocks? Can this be used for load bearing walls of a building? What are the advantages and disadvantages?**

Answer:

Hollow concrete blocks are precast masonry units made from cement, sand, aggregate and water, with one or more hollow cores. They are commonly used for partition walls and, when properly designed, can also be used for structural/load-bearing masonry.



The main characteristics of hollow concrete blocks are as follows:

- i. **Lightweight:** The hollow cores reduce the weight compared with solid concrete blocks or bricks.
- ii. **Uniform size and shape:** They are manufactured to standard dimensions, giving better dimensional accuracy.
- iii. **Hollow cores:** The cavities reduce dead weight and can also accommodate reinforcement, grout, or services.
- iv. **Good compressive strength:** The strength depends on the mix, manufacturing method and curing.
- v. **Low water absorption:** Properly manufactured blocks have relatively low water absorption.
- vi. **Good fire resistance:** Concrete is non-combustible and provides good resistance to fire.
- vii. **Good thermal and sound insulation:** The air cavities provide some insulation against heat and sound.
- viii. **Durable:** Properly manufactured and cured blocks can have good resistance to weathering and deterioration.

Can hollow concrete blocks be used for load-bearing walls?

Yes, hollow concrete blocks can be used for load-bearing walls, provided that they have adequate compressive strength and are designed according to the applicable masonry/building code.

For load-bearing construction:

- Blocks of adequate strength and quality should be selected.
- Proper mortar should be used.
- Wall thickness must be sufficient for the applied loads.
- Openings, connections and lateral loads must be properly considered.

- For earthquake-prone areas such as Nepal, reinforced/grouted concrete-block masonry with proper horizontal and vertical reinforcement and structural detailing is preferable to relying on unreinforced hollow-block masonry.

Advantages

- Reduces dead load of the building.
- Requires less mortar and can provide faster construction.
- Provides reasonable thermal and acoustic insulation.
- Good fire resistance and durability.
- Requires relatively less maintenance.
- Can reduce construction cost when locally manufactured and properly used.

Disadvantages

- Generally, it has lower strength than solid concrete blocks of the same overall dimensions.
- The thin webs can be damaged during handling and construction.
- Poor-quality blocks may have high water absorption and low durability.
- Unreinforced hollow-block masonry has poor resistance to earthquake and lateral loads compared with properly reinforced masonry.
- Requires careful workmanship at joints, corners, openings and connections.

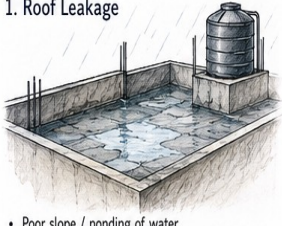
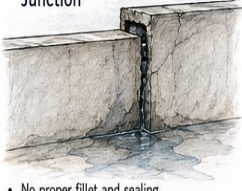
2. Describe with sketch common water proofing problems of a building in Nepal and their remedial measures taken.

Answer:

Waterproofing is essential for buildings in Nepal because of monsoon rainfall, freeze–thaw conditions in mountainous areas, fluctuating groundwater levels, poor drainage, and inadequate construction detailing.

Common problems and their remedies are as follows.

COMMON WATER PROOFING PROBLEMS OF A BUILDING IN NEPAL

1. Roof Leakage  • Poor slope / ponding of water • Cracks in concrete • Deteriorated or no waterproofing layer	2. Leakage at Parapet Wall Junction  • No proper fillet and sealing • Deterioration of waterproofing at junctions	3. Leakage Around Pipe Penetrations  • Improper sealing around pipes • Gaps due to movement or shrinkage	4. Leakage Around Drain Outlets  • Weak or missing waterproofing around drain • Poor workmanship
5. External Wall Dampness / Seepage  • Porous plaster / brickwork • No external waterproofing treatment • Rainwater penetration	6. Leakage at Window Sills  • No proper slope in sill • Gaps and poor sealing	7. Dampness Due to Rising Moisture  • Lack of damp proof course (DPC) • Capillary action from ground	8. Leakage in Bathroom / Wet Areas  • Improper waterproofing under tiles • Cracks in tiles and joints • Poor slope to drain
9. Expansion / Construction Joint Leakage  • Missing or damaged waterstop • Improper sealing of joints	10. Planter Box Leakage  • No waterproofing inside planter • Soil moisture causing seepage	11. Cracks in Walls / Slabs  • Structural or shrinkage cracks • Water penetration through cracks	12. Deterioration of Waterproofing Layer  • Aging of materials • UV exposure, poor quality materials and workmanship

Note: Regular inspection, proper design, quality materials and good workmanship are essential to prevent water leakage problems.

i. Dampness rising from the ground:

Moisture rises through foundations and walls by capillary action when an effective Damp-Proof Course (DPC) is absent or damaged.

Symptoms: Damp patches, peeling paint, salt deposits and deterioration of plaster near the floor.

Remedial measures:

- Provide or repair the DPC at plinth level.
- Inject a suitable chemical damp-proofing compound into existing walls.
- Provide a plinth protection and proper surface drainage around the building.
- Repair damaged plaster and apply waterproof coating after the wall has dried.

ii. Leakage through roof/terrace:

Cracks in the roof slab, defective waterproofing layers, inadequate slope or blocked outlets allow rainwater to penetrate the building.

Remedial measures:

- Provide a proper roof slope toward rainwater outlets.
- Repair cracks using suitable sealants or polymer-modified mortar.
- Provide waterproofing membrane/coating over the roof.

iii. Leakage through walls:

Rainwater penetrates external walls through cracks, porous masonry, defective joints and poorly sealed openings.

Remedial measures:

- Seal cracks and defective joints.
- Apply suitable exterior waterproof coating.
- Repair damaged pointing/plaster.
- Provide weatherproof flashing above doors and windows.

iv. **Basement and foundation leakage:**

Groundwater can enter through cracks in basement walls, construction joints and floor slabs, particularly where the groundwater table is high.

Remedial measures:

- Provide external waterproofing membrane where accessible.
- Seal cracks using suitable injection grouting.
- Provide a drainage layer and perforated drain around the foundation where appropriate.
- Use a sump and pump system where groundwater cannot be drained naturally.

v. **Leakage around doors and windows:**

Improper sealing between the frame and wall permits rainwater to enter.

Remedial measures:

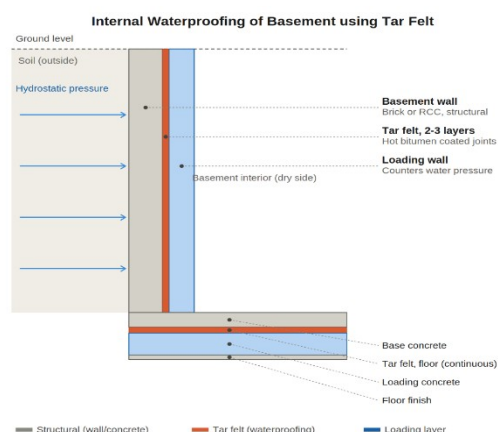
- Seal joints using waterproof sealant.
- Provide drip moulds and adequate sill projections.
- Provide proper flashing above openings.
- Ensure window sills have an outward slope.

vi. **Leakage from toilets and bathrooms:**

Water penetrates through floors and walls due to defective waterproofing, cracks, pipe penetrations and improperly sealed joints.

Remedial measures:

- Provide waterproofing membrane below the floor finish and upturn it along walls.
- Provide proper floor slope toward the floor drain.
- Repair damaged tiles and grout.
- Test the waterproofing before laying the final floor finish.



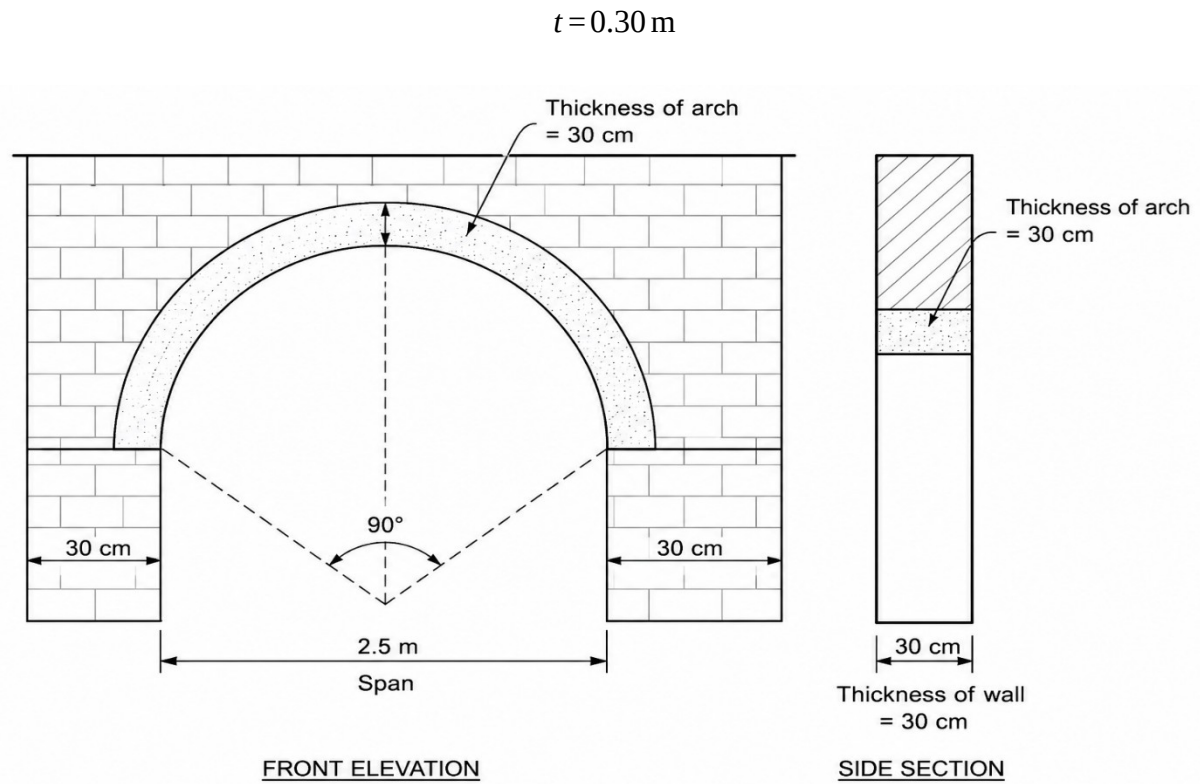
3. a. An arch of 2.5 m span subtends an angle of 90° at the center. The thickness of arch is 30 cm and thickness of wall is also 30 cm. Calculate the quantity of arch masonry work.

Answer:

Given,

$$L = 2.5 \text{ m},$$

$$\theta = 90^\circ,$$



For a circular arch,

$$L = 2R \sin \left(\frac{\theta}{2} \right)$$

$$2.5 = 2R \sin 45^\circ$$

$$R = \frac{2.5}{2(0.7071)}$$

$$R = 1.768 \text{ m}$$

Length of the arch is given by:

$$l = R\theta$$

Since,

$$90^\circ = \frac{\pi}{2} \text{ radians}$$

$$l = 1.768 \times \frac{\pi}{2}$$

$$l = 2.777 \text{ m}$$

Quantity of arch masonry:

$$V = l \times t \times t_w$$

where,

$$t = 0.30 \text{ m},$$

$$t_w = 0.30 \text{ m}$$

Hence,

$$V = 2.777 \times 0.30 \times 0.30$$

$$V = 0.24993 \text{ m}^3$$

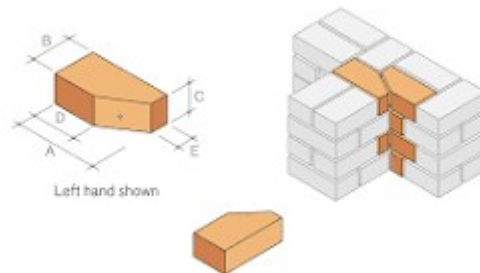
$$\therefore \text{Quantity of arch masonry} \approx 0.25 \text{ m}^3$$

4. What do you mean by a Kingstone?

Answer:

A Kingstone is a special wedge-shaped stone or brick used in the construction of an arch, generally placed near the springing or at a specified position to help maintain the proper arrangement and stability of the arch voussoirs.

It is properly cut to fit between adjacent voussoirs and helps in transferring the compressive forces through the arch.



5. **What do you know about roof truss? With sketch give the types, and properties of truss used in Nepal. What types of loads are considered in designing of roof truss?**

Answer:

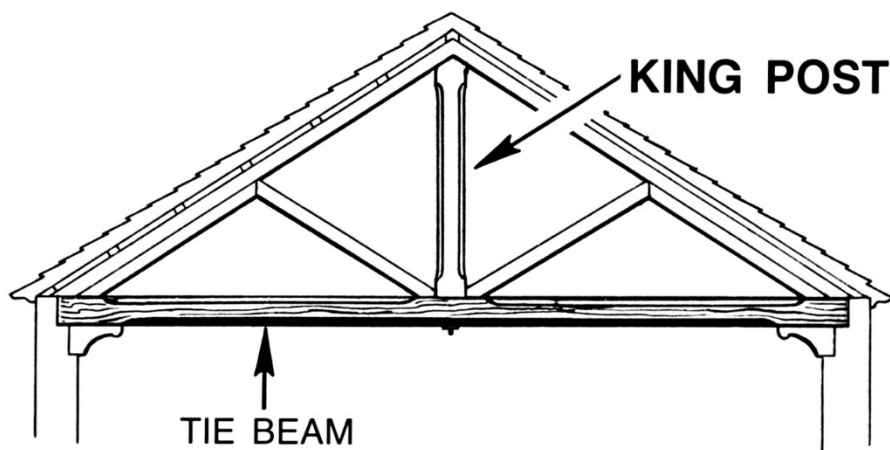
A roof truss is a framework of straight structural members arranged in a triangular or interconnected pattern to support the roof covering and transfer the loads safely to the supporting walls or columns. The members are mainly subjected to axial tension and compression.

When the span of roof exceeds 5 m and where there are no inside walls to support the purlins, trusses are provided at suitable interval along the length of the room. Spacing of truss is generally limited to 3 m for wooden trusses.

Types of trusses used in Nepal:

i. King-post truss:

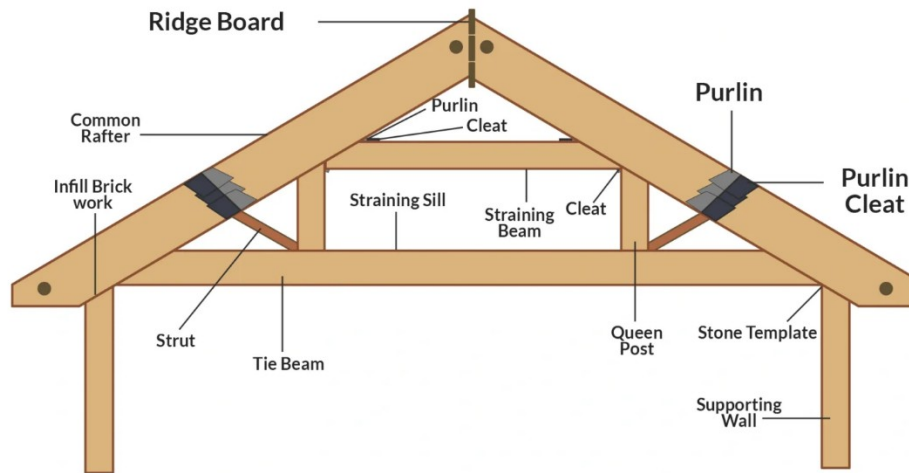
- It consists of framework comprising of two principle rafters, one tie beam, two inclined struts and a king post.
- The trusses are spaced at a distance of not more than 3 m c/c.
- It is commonly suitable for small spans, used for spans 9 to 12 m.



ii. Queen-post truss:

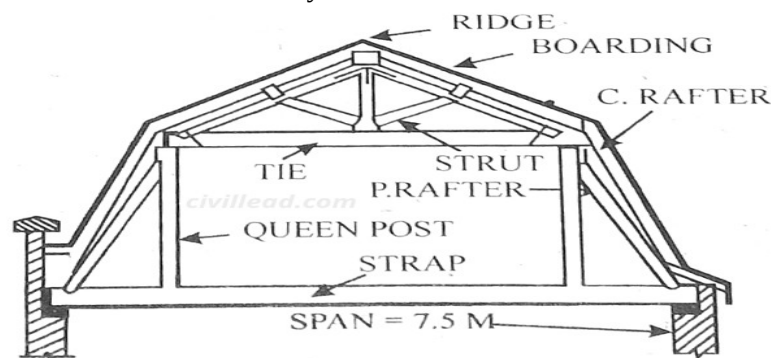
- It consists of framework comprising of two principal rafters, two queen posts and one straining beam.
- It differs from king truss in having two vertical posts and the vertical posts are known as queen posts.
- Top of the queen posts are connected by a horizontal piece known as straining beams.

- They are the tension members used for the span upto 14 m.



iii. Mansard roof truss:

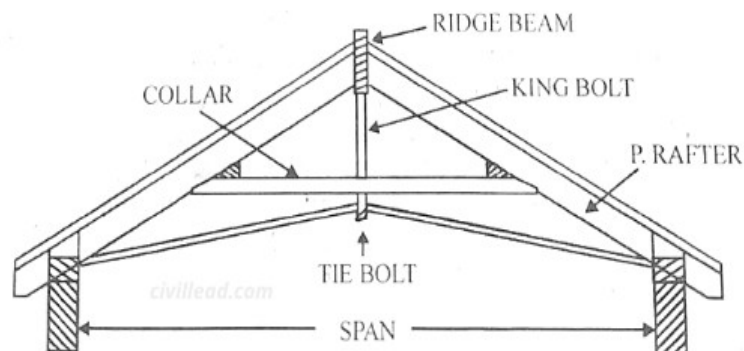
- It has two pitches, the lower varying from 60 to 70 degree and upper from 30 to 40 degree.
- This truss is combination of kingpost and queen post truss.
- It is a two-storey truss and upper portion consisting of king post truss and lower portion queen post truss.
- This type of truss provided economy in space and cost but because of its ungainly appearance its use become outdated these days.



Mansard Truss Roof

iv. Composite roof truss:

- Roof truss made up of two materials such as timber and steel is known as composite roof truss.
- In composite roof truss, the tension members are made of steel while compression members are made of timber.

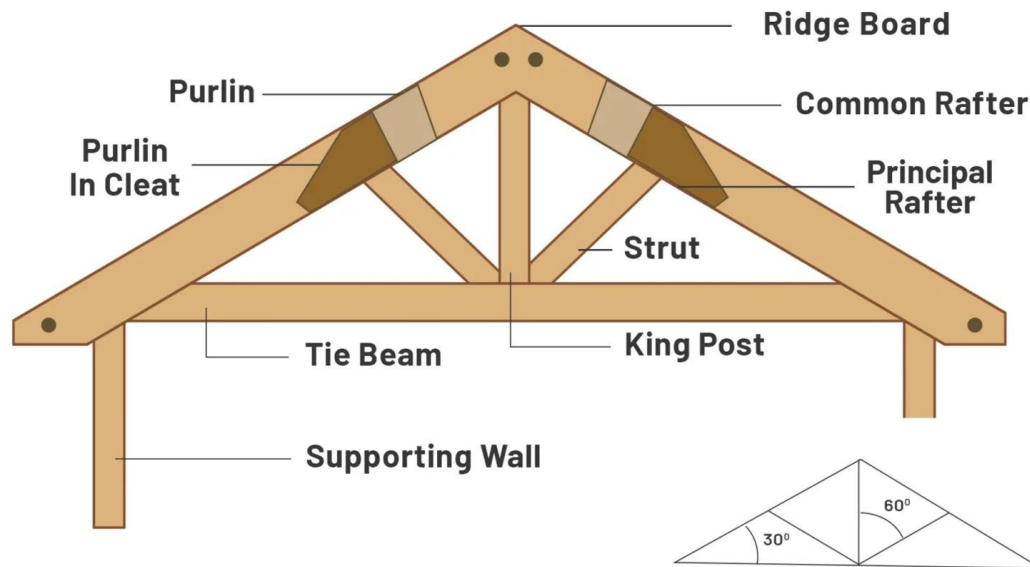


Composite Truss Roof

v. Combination of king-post and queen-post truss:

It is constructed by combination of king-post and queen-post, which is suitable for the span up to 18 m.

In Nepal, steel trusses made from angle sections, hollow structural sections or tubular members are commonly used for larger roofs such as industrial buildings, halls, warehouses, schools and other public buildings.



The important properties of steel truss are:

- i. **Lightweight:** Steel trusses provide high strength with relatively low self-weight.
- ii. **Economical:** They are suitable for covering moderate to large spans without intermediate supports.
- iii. **Prefabrication:** Steel trusses can be fabricated in workshops and assembled quickly at site.
- iv. **Suitable for pitched roofs:** They provide the required slope for drainage of heavy monsoon rainfall.
- v. **Durability:** Properly protected steel trusses have a long service life.
- vi. **Earthquake consideration:** Because Nepal is highly seismic, roof trusses should be light, properly anchored and adequately braced to prevent displacement during earthquakes.
- vii. **Local availability:** Timber trusses may be used where suitable local timber is available, while steel trusses are common for larger spans.
- viii. **Ease of maintenance:** Steel trusses require protection against corrosion, while timber requires protection against decay and termites.

The following types of loads are generally considered in a roof truss:

- i. Dead load:

Includes the self-weight of truss members, roof covering, purlins, ceiling, insulation, bracing and other permanent fixtures.

ii. Live load:

Includes temporary loads due to:

- Workers during construction and maintenance
- Maintenance equipment
- Other temporary loads on the roofs

iii. Wind load

Wind pressure or suction acting on the roof is particularly important for lightweight roof trusses.

iv. Snow load

Snow load should be considered in snow-prone areas of Nepal, particularly in high-altitude mountainous regions.

v. Seismic load:

For buildings in Nepal, earthquake effects and the associated inertial forces should be considered, particularly for the connections, bracing and anchorage of the roof truss.

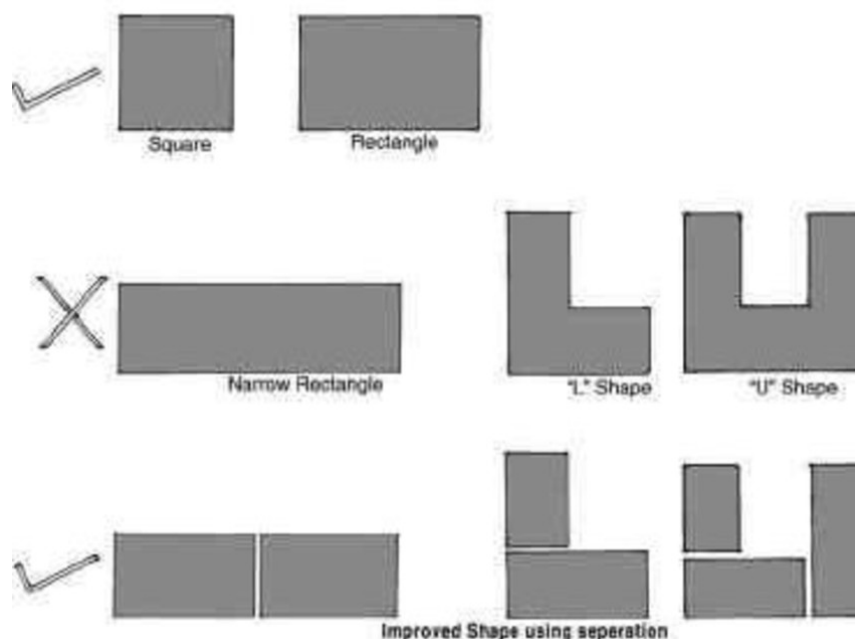
6. **What is the principle of earthquake design of buildings? Is it practical and normal to design a building earthquake proof?**

Answer:

The core philosophy of seismic design is not to make a building completely rigid, but to ensure it is ductile and regular so that it can absorb and dissipate the massive dynamic energy of an earthquake without collapsing. The principle of earthquake design of buildings is given below:

A. Symmetry, Shape, and Proportion:

- Buildings must have simple, regular, and symmetrical geometries in both plan and elevation. Structures with reentrant corners (e.g., L, U, H, or V-shaped layouts) suffer severe stress concentrations during seismic shaking.
- The breadth-to-length ratio of the building's overall plan as well as any individual room enclosed by load-bearing walls, should not exceed 1:3.
- The total height of the building should not exceed 3 times its minimum width.
- Excessively slender buildings experience high compressive forces that degrade the strength and stiffness of their columns.



B. Structural Regularity:

To prevent seismic forces from concentrating in a single vulnerable area, structures must avoid sudden vertical and horizontal changes. The two most critical vertical irregularities are:

- **Stiffness Irregularity (Soft Storey):** This occurs when a story's lateral stiffness is less than 70% of the story above, or less than 80% of the average stiffness of the three storeys above. In Nepal, this commonly occurs when the ground floor is left open for parking or shops, creating a high-risk structural "fuse".
- **Mass Irregularity:** This exists when the seismic weight of any single floor is greater than 200% of the adjacent floor's weight (e.g., placing heavy machinery or large water tanks at the roof level). This increases torsional vibrations, story shear, and overturning moments.

C. Structural Ductility and Detailing:

- Plain concrete is brittle and is prone to sudden failure under dynamic loads.
- Ductile detailing involves placing specific reinforcement steel configurations in beams, columns, joints, and shear walls.
- This allows the structural members to undergo large, controlled, inelastic deformations to dissipate earthquake energy uniformly.
- This does not prevent damage, but it prevents sudden collapse and buys critical time for occupants to safely evacuate.

D. Control and Alignment of Openings:

Large openings weaken load-bearing walls.

- **Size Restriction:** The total area of door and window openings in a load-bearing wall must be less than one-third (33.3%) of the total wall area.
- **Vertical Alignment:** Openings across different floors must be aligned vertically one above the other to ensure a continuous and uninterrupted path for load transfer to the foundation.

E. Traditional Concepts:

Long before modern building codes, traditional Nepalese structures integrated indigenous seismic-resistant features:

- **Horizontal Timber Bands:** Placing timber bands at the plinth, lintel, and roof levels to tie the walls together and prevent lateral separation.
- **Interlocking Joints & Wooden Pegs:** Utilizing flexible wood-to-wood connections instead of rigid joints to absorb and damp seismic vibrations.
- **Timber Frames with Masonry Infill:** Using flexible wood frames to carry the structural loads while masonry acts only as a ductile infill.
- **Lightweight Roofs:** Utilizing light materials (such as slate or CGI sheets) to minimize the inertial mass at the top of the building.

No building can realistically be guaranteed to be completely “earthquake-proof.” Earthquakes vary greatly in magnitude, duration, frequency content, distance from the fault, and ground conditions. Designing a structure to remain completely undamaged under every possible earthquake would be extremely expensive and generally impractical.

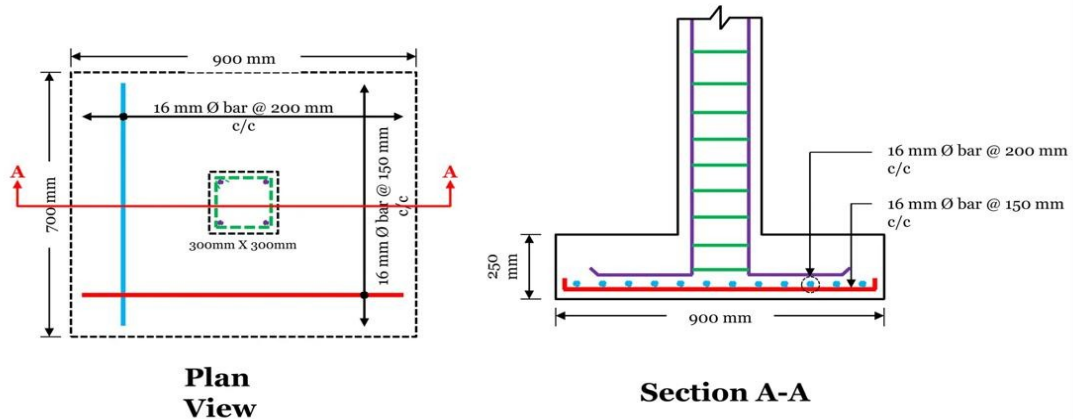
Therefore, the normal engineering approach is to design earthquake-resistant buildings, rather than absolutely earthquake-proof buildings.

7. **a. Describe with sketch design steps of a foundation footings of a building structure.**
b. Name types of footings of common buildings as practiced in Nepal.

Answer:

a. The design of a building footing involves determining the size, depth, thickness, reinforcement, and safety against shear, bending, and settlement. For an RCC isolated footing, the general design procedure is as follows:

Example



- i. Collection of design data:

The information such as column size, shape, loads, safe bearing capacity of the soil, unit weight of soil and concrete, properties of concrete and reinforcement etc. is required to be collected.

- ii. Determination of the required area of footing:

The required footing area is obtained from the service load and allowable bearing capacity of soil:

$$A_f = \frac{P}{q_{allowable}}$$

where,

A_f = Required area of footing

P = service load on column + Appropriate footing loads

$q_{allowable}$ = Allowable bearing capacity of soil

For square footing, $B = \sqrt{A_f}$

where B is the footing width

- iii. Check the soil pressure:

The actual soil pressure should not exceed the allowable bearing capacity:

$$Q = \frac{P}{BL} \leq q_{allowable}$$

For concentric loading, the pressure distribution is approximately uniform. For eccentric loading, the maximum and minimum pressures should be checked:

$$q_{max, min} = \frac{P}{BL} \left(1 \pm \frac{6e}{B} \right)$$

The footing should be proportioned so that excessive soil pressure and tensile contact pressure are avoided.

- iv. Determine the depth of footing:

The effective depth of the footing is selected based mainly on one-way shear and punching shear.

The critical sections for shear are checked at appropriate distances from the column face according to the adopted design code.

- v. Check for one-way shear:

The footing is treated as a beam projecting from the column.

The shear force at the critical section is calculated and checked against the permissible/design shear strength:

$$V_u \leq V_c$$

If the shear stress is excessive, the footing depth is increased.

- vi. Check for punching shear:

Punching shear is checked around the column at the prescribed critical perimeter.

The punching shear stress is:

$$v_u = \frac{V_u}{b_0 d}$$

where,

V_u = Punching shear force

b_0 = Critical perimeter

D = effective depth

The footing must satisfy: $v_u \leq v_c$

- vii. Calculate bending moment:

The footing projections beyond the column are treated as cantilevers carrying upward soil pressure. For a strip of footing, the bending moment is calculated at the column face.

For a uniformly loaded projection:

$$M_u = \frac{q_u l^2}{2}$$

Where, q_u is the factored soil pressure.

- viii. Design of the reinforcement

The required area of tensile reinforcement is calculated from the design bending moment.

For an approximate singly reinforced section:

$$A_{st} = \frac{M_u}{F_y j d}$$

Where, A_{st} is the required tensile reinforcement.

Reinforcement is generally provided near the bottom of the footing, in both directions for an isolated square footing.

- ix. Check minimum reinforcement and spacing

The provided reinforcement must satisfy:

- Minimum reinforcement requirements.
- Maximum permissible spacing.
- Adequate concrete cover.
- Proper anchorage and development length.

x. Check development length and column-footing connection:

The column bars must be adequately anchored into the footing, and the footing reinforcement must have sufficient development length.

The column-footing joint should be detailed properly to transfer axial load, shear, and bending moments.

xi. Check settlement and stability

The footing should be checked for:

- Total and differential settlement.
- Sliding.
- Overturning, where applicable.
- Bearing capacity failure.

xii. Reinforcement detailing:

Finally, prepare the plan and sectional drawing showing:

- Footing dimensions B×L.
- Footing thickness.
- Column dimensions.
- Bottom reinforcement in both directions.
- Column bars and anchorage.
- Foundation depth below ground level.

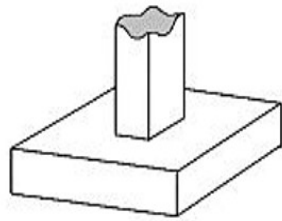
b. . Foundation is the lowermost part of structure which receives load from superstructure and transfers it to stable soil layer. The principal loads are:

1. **Dead Load:**
2. **Live (Imposed) Load:**
3. **Wind Load:**
4. **Earthquake (Seismic) Load:**
5. **Impact and Vibratory Loads:**

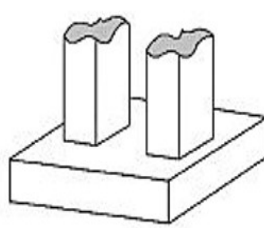
The types of footings of common building practiced in Nepal are as follows:

a. Shallow foundations:

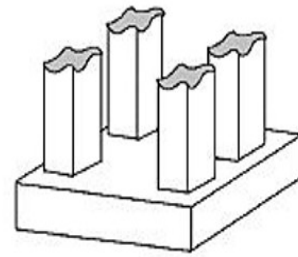
- Used when the soil with adequate bearing capacity is available near the ground surface (generally depth ≤ width).



Isolated Footing



Continuous footing



Mat footing

Types:

a. Strip (Wall) Footing:

- The footing that runs along the wall i.e., wall footing or continuous footing is known as strip footing.
- It is the slab supporting wall.

b. Isolated (Pad) Footing:

- Isolated footing is the slab supporting columns.
- It is constructed in RCC framed residential buildings.

c. Combined Footing:

- It is the slab supporting two or more columns in a line.
- Combined footing is provided to support columns at property line and closely spaced columns.

d. Strap (Cantilever) Footing:

- Strap footing is provided with a beam at foundation level to connect two columns which is known as strap beam.
- It is provided to support the column at property line.

e. Raft (Mat) Foundation:

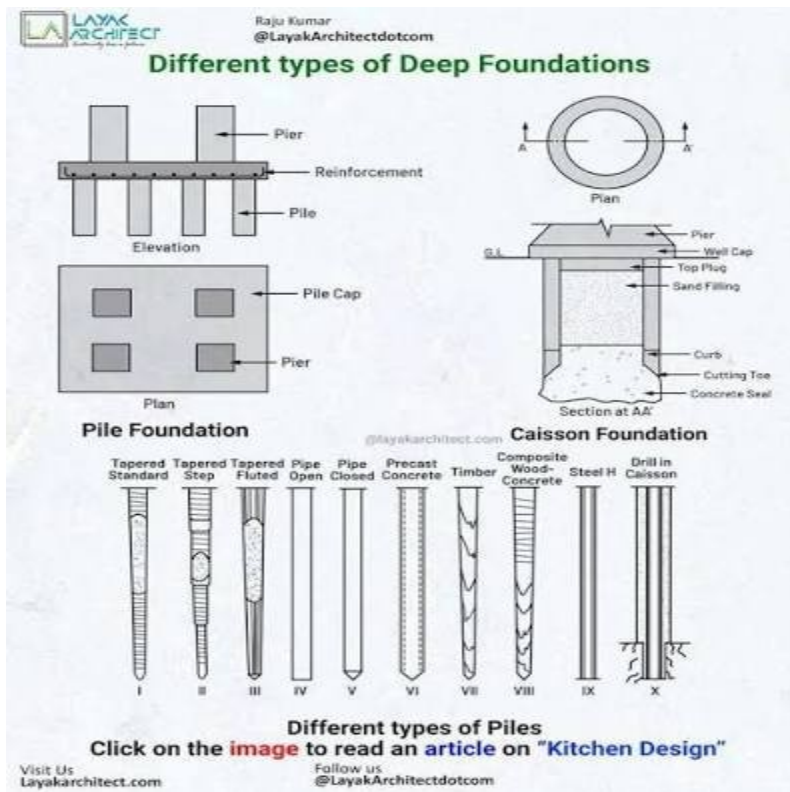
It is the slab supporting large number of columns in two or more rows.

It is used in compressible soil layer where isolated footings don't perform well and commercial buildings where isolated footing covers more than 50% of the plinth area.

Applications: Residential buildings, low-rise structures, and light to moderate loads.

b. Deep foundations:

Used when the upper soil is weak and stronger strata are available at greater depths (depth > width).



Types:

a. Pile Foundation:

- Pile foundation consists of long slender member called pile that transfer the load to the ground through bearing and skin friction.
- Piles also resist the structures against uplift and provide structural stability against lateral and overturning forces.

b. Pier Foundation:

- Pier foundations are those foundations which transfer loads that cannot be carried by shallow foundations.
- The foundation transfers the load to the ground only by bearing.
- When the top most soil is stiff clay, it resists the driving of bearing pile and hence in that case, pier foundation is used.

c. Caisson (Well) Foundation

- Well or caisson foundation is the watertight structure used in bridge pier, dam construction etc.
- It is a readymade hollow cylinder depressed into the soil up to the desired level and then filled with concrete that converts into foundations.

Applications: High-rise buildings, waterfront structures, and heavy industrial facilities.

8. **a. Describe the present and past practices in building construction in different zones of Nepal.**
b. Drawing a plan and vertical section of a storage building constructed in Jumla showing all elements and details.

Answer:

a. Nepal has considerable geographical and climatic variation, so traditional building practices developed according to local materials, climate, topography, culture, and available construction skills. Broadly, Nepal can be divided into Terai, Hill, and Mountain zones for discussing vernacular building practices.

i. Terai Region:

Past/traditional practices:

- Houses were generally single-storey and lightweight.
- Walls were commonly made of mud, bamboo, timber, wattle-and-daub, or mud-bonded brick.
- Roofs were constructed using thatch, clay tiles or timber.
- Raised plinths/platforms were used in flood-prone areas. In some area, houses were raised about 1–3 m above the surrounding ground.
- Wide roof overhangs helped protect walls from monsoon rain and provided shade.

Present practices:

- Traditional mud, bamboo and thatch houses are increasingly being replaced by burnt-brick masonry and RCC-framed buildings.
- Cement mortar, reinforced concrete slabs, steel reinforcement and GI sheets are widely used.
- In urban and semi-urban Terai, multi-storey RCC buildings are increasingly common.
- Modern construction provides greater durability and services but can be less climate-responsive when traditional ventilation and shading principles are ignored.

ii. Hill Region:**Past/traditional practices:**

- Stone masonry was the dominant construction method because stone was locally available.
- Mud mortar was commonly used for stone or brick masonry.
- Timber was extensively used for beams, columns, floors, doors, windows and roof structures.
- Roofs were commonly sloping, covered with slate, tiles, shingles or thatch.
- In Kathmandu Valley, traditional buildings particularly used brick, mud mortar and elaborately crafted timber elements.

Present practices:

- RCC framed structures with brick infill are now common in towns and cities.
- Cement mortar has largely replaced mud mortar.
- Reinforced concrete floors and roofs have become common.
- Earthquake-resistant features such as bands, proper reinforcement, vertical reinforcement, confined masonry and improved connections are increasingly incorporated.

iii. Mountain/Himalayan Region:**Past/traditional practices:**

- Buildings were constructed mainly from locally available stone and timber.
- Thick stone walls provided protection against the severe cold.
- Houses were generally compact, low-rise and rectangular.
- Roofs were steeply pitched and constructed using stone slabs, timber shingles or other locally available materials.
- In some traditional buildings, timber elements were incorporated with stone masonry, providing flexibility and structural integrity.

Present practices:

- Stone masonry continues to be important because transportation of construction materials to remote mountainous areas is difficult and expensive.
- Cement, steel, CGI sheets and RCC are increasingly used where transportation is possible.
- Modern buildings increasingly incorporate improved seismic detailing and stronger connections.
- Local materials and traditional construction techniques are still important, particularly in remote settlements.

b. A storage building constructed in Jumla consists of following elements:

i. **Foundation and plinth:**

- Rubble stone masonry foundation is generally used because stone is locally available in Jumla.
- The plinth is raised above ground to protect the building from moisture, snow and surface water.

ii. **Walls:**

- Thick stone masonry walls provide strength and thermal insulation.
- Mud mortar may traditionally be used, although cement mortar is common in modern construction.
- Small openings are preferred to reduce heat loss in the cold climate.

iii. **Floor:**

- Timber beams/joists support the floor.
- A raised storage platform may be provided to keep stored grains and materials away from dampness and pests.

iv. **Roof:**

- A steeply sloping timber roof is appropriate for Jumla's snowy and cold climate.
- Timber rafters support the roof covering.
- Adequate roof overhang protects the masonry walls from rain and snow.

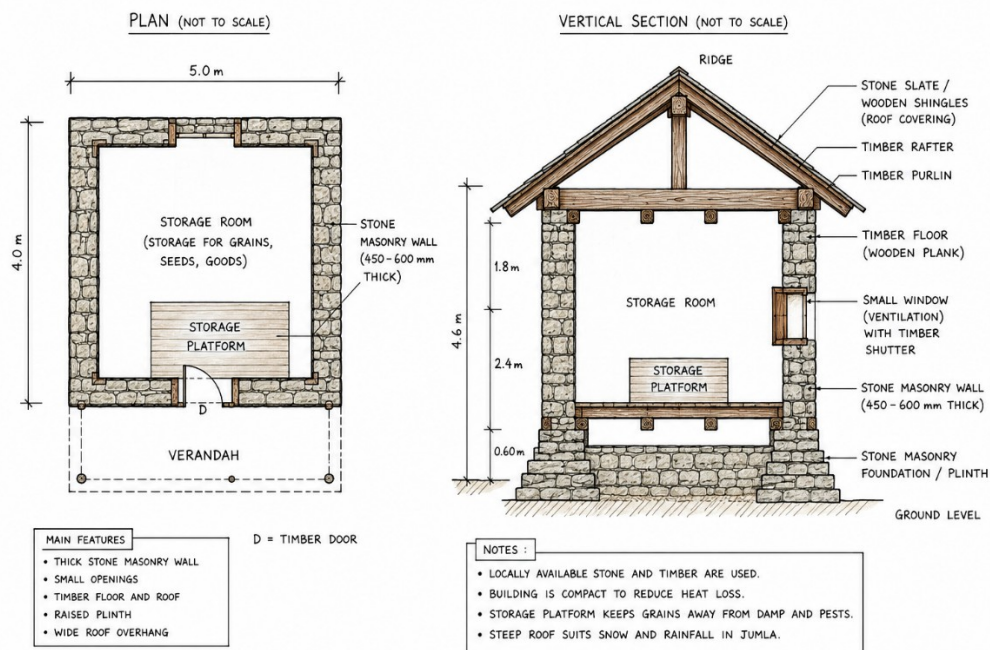
v. **Openings:**

- Doors and windows are generally kept relatively small.
- Timber frames and shutters provide insulation and security.

Functional considerations

- The storage space should be dry, cool, well ventilated and protected from rodents.
- The building is generally compact to reduce heat loss.
- Local stone and timber reduce transportation requirements and suit the Himalayan environment.

STORAGE BUILDING – JUMLA



9. **What are the general ethics of architects in professional practice? Illustrate the provisions being followed internationally and that included in Nepal acts.**

Answer:

Professional ethics are the moral principles and standards that guide an architect's conduct while dealing with the public, client, profession, colleagues, contractors and other professionals. The main purpose is to ensure that architectural services are provided with honesty, competence, independence, fairness, confidentiality and responsibility.

The general ethics of architects in professional practice:

- i. Maintain honesty and integrity in all professional activities.
- ii. Protect public health, safety and welfare through responsible design and supervision.
- iii. Maintain professional competence and undertake only work for which he/she has adequate knowledge and experience.
- iv. Exercise independent and impartial professional judgment rather than allowing personal or financial interests to influence decisions.
- v. Avoid discrimination on grounds such as gender, race, religion, caste, disability, etc.
- vi. Respect intellectual property and copyright and give proper recognition to the work and ideas of others.
- vii. Charge fair and transparent professional fees and avoid improper financial benefits, commissions or inducements.
- viii. Protect the environment and promote sustainable and socially responsible architecture.
- ix. Maintain and improve professional knowledge through continuing learning and professional development.

Nepal does not currently have a separate Architects Act comparable to a standalone Architects Council Act. Architectural practice falls within the statutory framework of the Nepal Engineering Council (NEC). NEC Act's schedule includes "Building and Architect" under Civil Engineering and also lists Architect Engineering among engineering professions.

The principal legal provisions are the Nepal Engineering Council Act, 2055 (1999) and the Nepal Engineering Council Regulations, 2057 (2001), as amended.

- i. Discipline and honesty

Under Rule 18 of the NEC Regulations, registered professionals must conduct their professional services in a disciplined, honest and integral manner, consistent with professional dignity and interests.

- ii. Confidentiality

Professional information relating to a client should not be disclosed to others without the client's approval, except where disclosure is required by law or to the appropriate authority.

- iii. Non-discrimination

Professionals must not discriminate against clients on the basis of religion, race, gender, caste or other such grounds while providing professional services.

- iv. Identification on professional documents

When signing maps, designs, specifications, estimates and other professional documents, the professional must clearly state the name, position/designation and registration number.

- v. Negligence and recklessness

The amended provisions specifically require professionals not to perform work with negligence or recklessness. The Act also states that engineering professionals must not act with ill intention or negligence/recklessness.

10. Explain the environmental issues in urban and rural development of our country and how those should be dealt with while designing a storage building of Nepal food corporation.

Answer:

Nepal has diverse geographical conditions, from the Terai to the hills and mountains, and therefore environmental problems differ between urban and rural areas. Government assessments identify air, water and solid-waste pollution as major urban concerns, while rural areas face soil erosion, landslides, floods, unsafe water, watershed degradation and forest depletion.

Major Environmental Issues in Urban Development:

The important environmental problems associated with urban development are:

- i. Air pollution due to vehicles, industries, construction activities and dust.
- ii. Water pollution from untreated domestic sewage, industrial effluent and surface runoff.
- iii. Solid-waste problems due to inadequate collection, segregation, treatment and disposal.
- iv. Noise pollution from traffic, construction and industries.
- v. Loss of open spaces and vegetation because of uncontrolled urbanization.
- vi. Flooding and drainage congestion due to excessive impervious surfaces and inadequate drainage.
- vii. Land and river encroachment and construction in environmentally sensitive areas.
- viii. High energy consumption and increasing greenhouse-gas emissions.
- ix. Urban heat-island effects caused by dense construction and reduced vegetation.
- x. Traffic congestion and associated environmental pollution.

Nepal's climate-risk assessment also identifies urban problems such as land, air, water and noise pollution, inadequate drainage, river encroachment and informal construction.

Major Environmental Issues in Rural Development:

The important rural environmental problems include:

- i. Deforestation and degradation of watersheds.
- ii. Soil erosion and landslides, especially in the hills and mountains.
- iii. Flooding and riverbank erosion.
- iv. Degradation of agricultural land.
- v. Scarcity and contamination of drinking water.
- vi. Poor sanitation and waste management.
- vii. Loss of biodiversity and natural habitats.
- viii. Unplanned settlement and road construction.
- ix. Forest and natural-resource depletion.
- x. Climate-change impacts such as changing rainfall, drought, floods, landslides, changing snowfall and extreme temperatures.

Environmental Considerations for a Nepal Food Corporation Storage Building

A Nepal Food Corporation storage building is particularly important because it stores food grains and other food commodities. Therefore, environmental design must protect both the stored food and the surrounding environment.

i. Site selection:

The site should:

- Be located away from flood-prone areas, landslide-prone slopes and unstable ground.
- Avoid environmentally sensitive areas, wetlands and river corridors.
- Have adequate road access for transportation without causing excessive disturbance.
- Provide sufficient space for drainage, loading/unloading, parking and landscaping.
- Consider prevailing wind direction so that dust, smoke and other pollutants do not enter the storage area.
- Provide adequate separation from potential sources of contamination.

ii. Protection against moisture and water:

Moisture is one of the major threats to stored food.

- The building should be constructed on a raised plinth.
- A damp-proof course should be provided.
- Floors should be impervious, durable and easy to clean.
- Roofs should be properly waterproofed.
- Rainwater should be collected and safely drained away.
- Surface drainage should be designed so that water does not accumulate around the foundation.

iii. Natural ventilation and lighting

Proper ventilation helps control temperature, humidity and condensation.

- Provide suitably located ventilators and windows.
- Use natural lighting wherever practical.
- Avoid direct sunlight falling excessively on stored food.
- Provide controlled mechanical ventilation where natural ventilation is insufficient.
- Openings should be protected against rodents, insects and birds.

iv. Temperature and humidity control

Food grains can deteriorate because of excessive temperature and moisture. The building should therefore have:

- Thermal insulation where necessary.
- Proper roof insulation.
- Moisture-resistant walls and floors.
- Adequate ventilation.
- Temperature and humidity monitoring.
- Suitable storage arrangement with sufficient clearance between food stacks and walls.

v. Solid-waste management:

Waste generated during storage, handling and cleaning should not be dumped openly.

Provision should be made for:

- Waste segregation.
- Covered waste containers.
- Safe disposal of damaged/spoiled food.
- Regular cleaning of the storage premises.
- Proper pest and rodent-control measures.

vi. Energy-efficient design

The storage building should minimize energy consumption through:

- Proper building orientation.
- Natural daylight and ventilation.
- Insulated roof and walls.
- Energy-efficient lighting and equipment.
- Energy-efficient refrigeration or ventilation systems where required.

11. You are assigned to design a storage building for Nepal Food Corporation in Nepalgunj. Write step by step procedure you follow to design such a building.

Answer:

A storage building for the Nepal Food Corporation should be designed to provide adequate storage capacity, protection of food grains from moisture, pests and temperature variations, efficient handling of materials, and structural safety. The following procedure may be followed:

i. Collection of Basic Information and Site Investigation

- Determine the type and quantity of food grains to be stored.
- Estimate the required storage capacity based on present and future demand.
- Collect information about the site location, available land and accessibility.
- Carry out a topographical survey and prepare the site plan.
- Conduct a geotechnical investigation to determine soil type, bearing capacity, groundwater level and settlement characteristics.

ii. Selection of Suitable Site

The site should:

- Have adequate land for the warehouse, roads, loading/unloading areas and future expansion.
- Be accessible by road and preferably connected to major transportation routes.
- Be above flood-prone levels and have proper natural or artificial drainage.
- Be located away from sources of contamination, excessive dust and fire hazards.

iii. Determination of Storage Requirements

Determine:

- Required storage capacity in tonnes.
- Duration of storage.
- Required clear height and floor area.
- Number and size of storage compartments/bins.

Adequate additional space should be provided for aisles, circulation, ventilation and handling operations.

iv. Preparation of Functional Layout

Prepare a functional layout showing:

- Main storage hall/warehouse.
- Receiving and unloading area.
- Loading area.
- Office and administrative spaces.
- Grain cleaning and processing areas, if required.
- Equipment and machinery rooms.
- Stores and maintenance areas.
- Toilets and other service facilities.
- Drainage and fire-fighting facilities.

The layout should minimize unnecessary movement of grain and vehicles.

v. Architectural Planning

Prepare the architectural design considering:

- Suitable building orientation.
- Adequate floor area and clear internal height.
- Large doors for trucks and material handling equipment.
- Proper ventilation and natural lighting.
- Easy cleaning and maintenance.

- Moisture-proof construction.
- Provision for future expansion.

For Nepalgunj's hot climate, proper ventilation, roof insulation and solar protection are important.

vi. Structural System Selection

Select a suitable structural system such as:

- Reinforced concrete framed structure,
- Steel portal frame,
- Steel truss with masonry/RCC walls, or
- A suitable combination of these.

The selection should depend on span, storage requirements, economy, construction materials, durability and availability of skilled labor.

vii. Structural Design

Design the structural components for all relevant loads, including:

- Dead load.
- Live load.
- Grain/storage load.
- Wind load.
- Earthquake load.
- Roof loads.
- Temperature effects where applicable.
- Equipment and machinery loads.

Design:

- Roof and roof trusses/portal frames.
- Beams and columns.
- Floor slab and foundation.
- Walls and other structural elements.
- Connections and bracing systems.

The design should comply with the applicable Nepal Building Code (NBC) and relevant standards.

viii. Foundation Design

Using the geotechnical investigation results:

- Select an appropriate foundation type.
- Determine allowable bearing capacity.
- Calculate foundation dimensions.
- Check bearing pressure and settlement.
- Check sliding, overturning and uplift where relevant.
- Provide adequate reinforcement and detailing.

The foundation should be designed considering the seismic condition and local soil characteristics of Nepalgunj.

ix. Floor Design:

The floor is a critical component because it receives grain and material-handling loads. It should:

- Have adequate load-carrying capacity.
- Be durable and abrasion-resistant.
- Be smooth and easy to clean.
- Resist moisture penetration.
- Have proper slope where required for cleaning and drainage.

x. Moisture and Damp-Proofing Measures

Since moisture can damage stored food grains, provide:

- Damp-proof course.
- Damp-proof flooring where necessary.
- Waterproof roof construction.
- Proper roof drainage.
- Raised floor/plinth above surrounding ground level.
- Proper sealing of openings against rainwater.